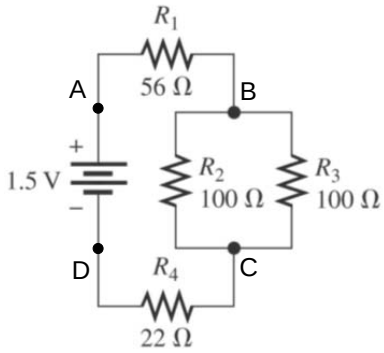
EE3404 Electric Circuit Theory Assignment One (Complete All 4 Questions)

Due: 30 September 2026

1. Series-Parallel Circuits (10 marks)



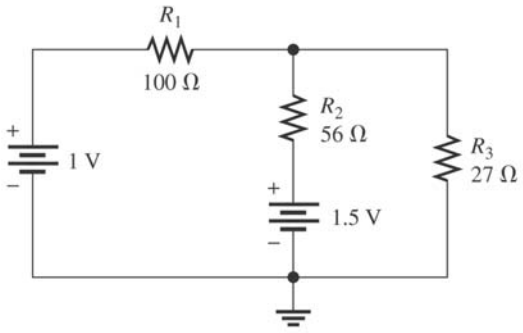
   (a) Determine the total resistance R_T of the circuit.
   (b) Determine the currents through the resistor R_1, R_2, R_3, and R_4 respectively.
   (c) Determine the voltage drops in AB, BC, CD, and AD respectively.

2. Voltage Dividers with Resistive Loads (10 marks)

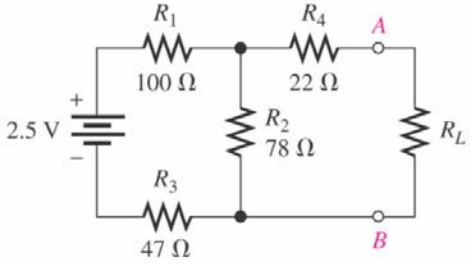
   A voltage divider consists of a 15 V source, and two resistors which are 330Ω and 220Ω respectively. Voltage output is taken across the 220Ω resistor.
   (a) Draw the circuit diagram for this voltage divider.
   (b) Calculate the unloaded output voltage.
   (c) What will be the output voltages if a load resistor of (i) 1.0 MΩ, and (ii) 1 kΩ is connected across the output?
   (d) Does the 10X design rule apply for (i) the 1.0 MΩ load, and (ii) the 1 kΩ load? Discuss with the values obtained in the steps above.

3. Superposition Theorem (10 marks)

   For the given circuit, use the superposition theorem to find the voltage drop across R_3.



4. Thevenin's Theorem (10 marks)



   (a) Find the Thevenin equivalent model as seen from terminals A and B for the circuit.
   (b) Use Ohm's law (I = V/R) and the power formula ($P = I^2R$) to find the load power, P_L, for each value of load resistance R_L: (i) 50Ω (ii) 73Ω and (iii) 100Ω.
   (c) Which of the above resistance values in (b) gives the greatest load power? What is its relationship with the Thevenin equivalent resistance of the circuit?